Wireless Networks

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What is a Wireless Local Area Network

- A Wireless Local Area Network (WLAN) is a type of local area network that uses high frequency radio waves rather than wires to communicate between network-enabled devices.
- **Wi-Fi,** a brand name given by the Wi-Fi Alliance (formerly Wireless Ethernet Compatibility Alliance), is a generic term that refers to the communication standard for the wireless network which works as a Local Area Network to operate without using the cable and any types of wiring.
- WIFI uses radio technology to transmit and receive data at high speed.

TYPES OF WI-FI

Standards	Year of Release	Description
Wi-Fi-1 (802.11b)	1999	This version has a link speed from 2Mb/s to 11 Mb/s over a 2.4 GHz frequency band
Wi-Fi-2 (802.11a)	1999	After a month of release previous version, 802.11a was released and it provide up to 54 Mb/s link speed over 5 Ghz band
Wi-Fi-3 (802.11g)	2003	In this version the speed was increased up to 54 to 108 Mb/s over 2.4 GHz
802.11i	2004	This is the same as 802.11g but only the security mechanism was increased in this version
802.11e	2004	This is also the same as 802.11g, only Voice over Wireless LAN and multimedia streaming are involved
Wi-Fi-4 (802.11n)	2009	This version supports both 2.4 GHz and 5 GHz radio frequency and it offers up to 72 to 600 Mb/s speed
Wi-Fi-5 (802.11ac)	2014	It supports a speed of 1733 Mb/s in the 5 GHz band

Elements of WIFI network

- A **wireless access point (AP)** is a hardware device that allows wireless communication devices, such as PDAs and mobile computers, to connect to a wireless network.
  - Usually, an AP connects into to a wired network, and provides a bridge for data communication between wireless and wired devices.
- **A Service Set Identifier (SSID)** is a configurable identification that allows wireless clients to communicate with an appropriate access point.
  - With proper configuration, only clients with correct SSID can communicate with the access points.
  - In effect, the SSID acts as a single shared password between access points and clients.

Benefits of Wireless

- Increased flexibility
- Increased productivity
- Reduced costs
- Ability to grow and adapt to changing requirements

Why Wireless network is insecure

- Channel: Broadcast ⇒ Eavesdropping, Jamming, Active attacks on protocols
- Mobility: Portable devices ⇒ Not physically secured
- Resources: Limited memory and processing resources ⇒ Need simpler security
- Accessibility: May be left unattended

WIFI security

- Wireless security is the prevention of unauthorized access or damage to computers using wireless networks.
- The most common types of wireless security are Wired Equivalent Privacy (WEP) and Wi-Fi Protected Access (WPA)

Wired Equivalent Privacy Protocol(WEP)

- Wired Equivalent Privacy (WEP) Protocol is a basic security feature in the IEEE 802.11 standard, intended to provide confidentiality over a wireless network by encrypting information sent over the network.
- The original 802.11 wireless standard relied upon a cryptographic protocol called wired equivalent privacy or WEP.
- WEP was meant to provide users privacy equivalent to that of a dedicated wire, that is, immunity to most eavesdropping and impersonation attacks.
- WEP uses an encryption key shared between the client and the access point. To authenticate a user, the access point sends a random number to the client, which the client encrypts using the shared key and returns to the access point.
- From that point on, the client and access point are authenticated and can communicate using their shared encryption key.
- A key-scheduling flaw has been discovered in WEP, so it is now considered as unsecured because a WEP key can be cracked in a few minutes with the aid of automated tools.

Therefore, WEP should not be used unless a more secure method is not available.

Wi-Fi Protected Access (WPA)

- The alternative to WEP is **WiFi Protected Access** or WPA, approved in 2003. The IEEE standard 802.11i is now known as WPA2, approved in 2004, and is an extension of WPA.
- First, WEP uses an encryption key that is unchanged until the user enters a new key at the client and access point.
- WPA has a key change approach, called **Temporal Key Integrity Program (TKIP)**, by which the encryption key is changed automatically on each packet.
- WPA employs the extensible authentication protocol (EAP) by which authentication can be done by password, token, certificate, or other mechanism.
- For small network (home) users, this probably still means a shared secret, which is not ideal. Users are prone to selecting weak keys, such as short numbers or pass phrases subject to a dictionary attack.

Wi-Fi Protected Access (WPA)

- **WPA2 (Wi-Fi Protected Access version 2)** was introduced in 2004. The most important improvement this 802.11i wireless security standard offered over its predecessor was the implementation of the Advanced Encryption Standard (AES).
- **WPA3 (Wi-Fi Protected Access version 3)** is the latest security protocol with top standards. WPA3 protects against dictionary attacks and uses Simultaneous Authentication of Equals handshake, which protects its network from attacks that could be possible with WPA2 in place. WPA3 is really good on public networks (say in a coffee place), because it automatically encrypts the connection without any need for additional credentials.

Wi-Fi Protected Access (WPA)

- The setup protocol for WPA and WPA2 is much more robust than that for WEP.
- Setup for WPA involves three protocol steps: authentication, a four-way handshake (to ensure that the client can generate cryptographic keys and to generate and install keys for both encryption and integrity on both ends), and an optional group key handshake (for multicast communication.)
- WPA and WPA2 address the security deficiencies known in WEP.

Wireless network security issues and threats

- The most common WLAN security threats include:
  - **Spoofing and session hijacking** - where the attacker gains access to network data and resources by assuming the identity of a valid user
  - **Eavesdropping -** where unauthorized third parties intercept the data being transmitted over the secure network
  - **Evil Twin Attacks** -an adversary gathers information about a public network access point, then sets up their system to impersonate it. The adversary uses a broadcast signal stronger than the one generated by the legitimate access point; then, unsuspecting users connect using the stronger signal.

Wireless network security issues and threats

- **Accidental Association**
  - A user intending to connect to one LAN may unintentionally lock on to a wireless access point from a neighboring network.
- **Identity Theft (MAC Spoofing)**
  - Identity theft (or MAC spoofing) occurs when a cracker is able to listen in on network traffic and identify the MAC address of a computer with network privileges.
  - A **MAC address** is a hardware identification number that uniquely identifies each device on a network. The **MAC address** is manufactured into every network card, such as an Ethernet card or Wi-Fi card, and therefore cannot be changed
  - Most wireless systems allow some kind of MAC filtering to only allow authorized computers with specific MAC IDs to gain access and utilize the network.
  - However, a number of programs exist that have network "sniffing" capabilities. Combine these programs with other software that allow a computer to pretend it has any MAC address that the cracker desires, and the cracker can easily get around that hurdle.

Wireless network security issues and threats

- **Malicious Association**
  - "Malicious associations" are when wireless devices can be actively made by crackers to connect to a company network through their cracking laptop instead of a company access point (AP).
  - . The idea behind this type of attack may not be to break into a VPN or other security measures. Most likely the cracker is just trying to take over the client at the Layer 2 level.

Wireless Network Threats

- **Man-in-the – Middle attack**
  - This attack involves persuading a user and an access point to believe that they are talking to each other when in fact the communication is going through an intermediate attacking device. Wireless networks are particularly vulnerable to such attacks.
- **Denial of Service**
  - A Denial-of-Service attack (DoS) occurs when an attacker continually bombards a targeted AP (Access Point) or network with bogus requests, premature successful connection messages, failure messages, and/or other commands with an aim to affect the availability of the network resources.
  - These cause legitimate users to not be able to get on the network and may even cause the network to crash.
  - These attacks rely on the abuse of protocols such as the Extensible Authentication Protocol (EAP).

Securing Wireless Access Points

- The main threat involving wireless access points is unauthorized access to the network.
- Securing Wireless Networks /Access points

1. Use encryption.

   Wireless routers are typically equipped with built-in encryption mechanisms for router-to-router traffic.
2. Use antivirus and antispyware software, and a firewall.
3. Turn off identifier broadcasting. If a network is configured so that authorized devices know the identity of routers, this capability can be disabled, so as to thwart attackers.
4. Allow only specific computers to access your wireless network. A router can be configured to only communicate with approved MAC addresses.

Securing Wireless Access Points

5. Change default passwords of networking devices
6. Protect/ Hide your Service Set Identifier (SSID). To prevent outsiders from easily accessing your network, avoid publicizing your SSID.
7. Keep your access point software patched and up to date. The manufacturer of your wireless access point will periodically release updates to and patches for a device's software and firmware. Be sure to check the manufacturer's website regularly for any updates or patches for your device.
8. Connect using a Virtual Private Network (VPN).

Securing Wireless Access Points

9. Keep your access point software patched and up to date
10. Install a firewall

NETWORK AUDITING

- Wireless network auditing is an important part of WLAN security policy.
- The network needs to be regularly audited for rouge hardware.
- In this method the network is scanned and mapped for all access points and WLAN nodes.
- Then this is compared with previous network map.
- Commonly available network mapping tools like **netstumbler** and **wavelan-tool** can be used to do this.
- Specialized tools such as **Airsnort** can be used for auditing the network for weak keys, key reuse and WEP security settings.
- These methods include the same tests as those carried out by hackers for breaking into the network

Components of IEEE 802.11

- Stations
- Access point (AP)
- Basic service set (BSS)
- Extended service set (ESS)
- Distribution system (DS)

Stations

- All components that can connect into a wireless medium in a network are referred to as stations (STA).
- All stations are equipped with wireless network interface controllers (WNICs).
- Wireless stations fall into two categories: wireless access points, and clients. Access points (APs).
- Normally wireless routers, are base stations for the wireless network.
- They transmit and receive radio frequencies for wireless enabled devices to communicate with.
- Wireless clients can be mobile devices such as laptops, personal digital assistants, IP phones and other smartphones, or non-portable devices such as desktop computers, printers, and workstations that are equipped with a wireless network interface.

Basic service set

- The basic service set (BSS) is a set of all stations that can communicate with each other at PHY layer.
- Every BSS has an identification (ID) called the BSSID, which is the MAC address of the access point servicing the BSS.
- There are two types of BSS: Independent BSS (also referred to as IBSS), and infrastructure BSS.
- An independent BSS (IBSS) is an ad hoc network that contains no access points, which means they cannot connect to any other basic service set. Independent basic service set

Distribution system

- A distribution system (DS) connects access points in an extended service set.
- The concept of a DS can be used to increase network coverage through roaming between cells.
- DS can be wired or wireless.
- Current wireless distribution systems are mostly based on WDS or MESH protocols, though other systems are in use.

IEEE Architecture

- **Extended Service Set (ESS)** – It is a set of all connected BSS.
- **Distribution System (DS)** – It connects access points in ESS.



